SANTHOSH V S

Aspiring Embedded Systems & IoT Engineer
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PROFILE SUMMARY

Final-year Electronics & Communication Engineering student with hands-on experience designing and deploying IoT-based embedded systems using ESP32, ESP8266, and STM32 platforms. Skilled in sensor integration, real-time monitoring, cloud data visualization, and wireless communication (GSM, GPS). Seeking an internship/entry-level role in Embedded Systems or IoT development.

EDUCATION

Knowledge Institute of Technology, Salem *Sep 2023 – May 2027*
B.E. in Electronics and Communication Engineering | CGPA: 7.743 (up to 6h Semester)

The Gugai Hr. Sec. School *Jun 2020 – Mar 2023*
Class 12th: 87.33%
Class 10th:NIL

TECHNICAL SKILLS

- **Programming Languages:** C, Java
- **EV & Embedded Tools:** STM32CubeIDE, STM32CubeMX, S32 Design Studio, TS Master,
- **Circuit Design & Simulation:** Proteus, KiCad, LTspice, TinkerCAD,MATLAB
- **Protocols:** UART,I2C,CAN,SPI,USB
- **Domains of Interest:** Embedded Systems, Internet of Things (IoT), Electric Vehicle Technology, Automotive Electronics

PROJECTS & RESEARCH EXPERIENCE

Smart Railway Monitoring System

- Built an IoT-based railway track monitoring system on ESP32 to detect track cracks and improve railway safety.
- Integrated an ultrasonic sensor for crack detection, a GPS module for location tracking, and an ESP32-CAM for live visual monitoring.
- Used a SIM800L GSM module to trigger automatic SMS alerts with location data on fault detection, enabling faster response.

Patient Health Monitoring System

- Developed an IoT-based patient health monitoring system on ESP8266 for real-time tracking of vital parameters.
- Integrated a MAX30100 sensor for heart rate and SpO2 monitoring and a DS18B20 sensor for body temperature.
- Streamed sensor data to the ThingSpeak cloud platform for live visualization and remote monitoring.

Electric Bike Development *Indian E-Bike Championship*

- Designed and developed a self-manufactured electric bike as part of a multidisciplinary engineering team for a national-level competition.
- Led electrical-domain work including wiring architecture, motor-controller integration, and motor mapping, applying embedded control concepts to a real hardware system.
- Represented the team at the Indian E-Bike Championship in Bhopal, competing against teams from across India.

INTERNSHIP & TRAINING EXPERIENCE

- **Basic Electronics** — EG Technologies Pvt. Ltd (1 Week): Hands-on training in electronic components and circuit fundamentals.
- **Fundamentals of Telecommunication** — BSNL, In-Plant Training (1 Week): Studied telecommunication infrastructure and switching systems.

ACHIEVEMENTS

- Top 5 Finish and Best Cost Report Award at the Indian E-Bike Championship, organized by Hindustan Motorsports at IES University, Bhopal, Madhya Pradesh.

LANGUAGES

English, Tamil